Modeling Wetland Photosynthetic Recovery with HLS Remote Sensing Data

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523D – Reproducible Remote Sensing Models

Restored wetlands are increasingly promoted as a nature-based climate solution, but it is still unclear how quickly they recover carbon uptake function
(Valach et al., 2021; Schuster et al., 2025)

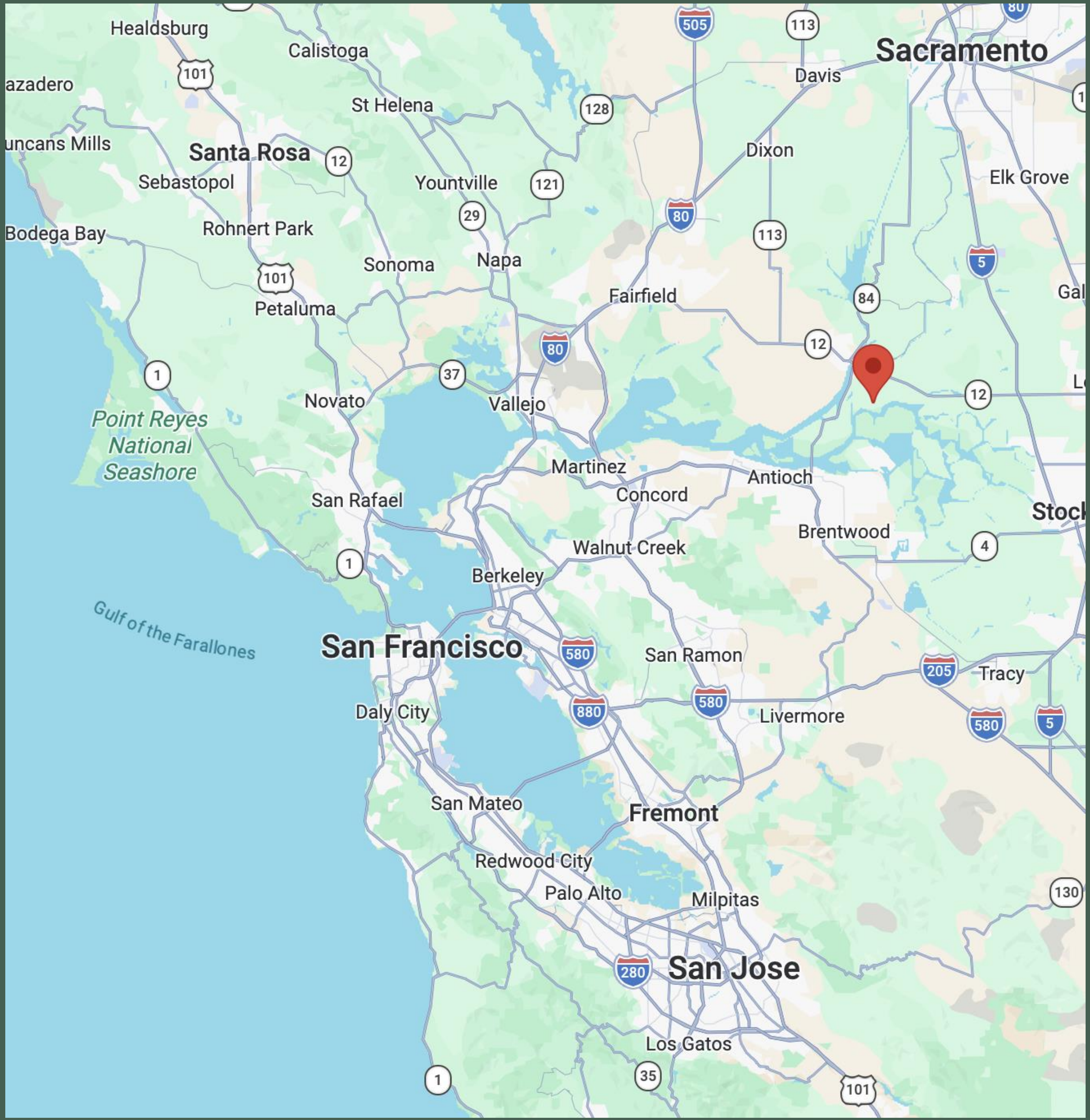
Understanding that trajectory could help land managers evaluate whether restoration is delivering expected climate benefits

Can we track and predict restored wetland photosynthetic function recovery using remote sensing data?

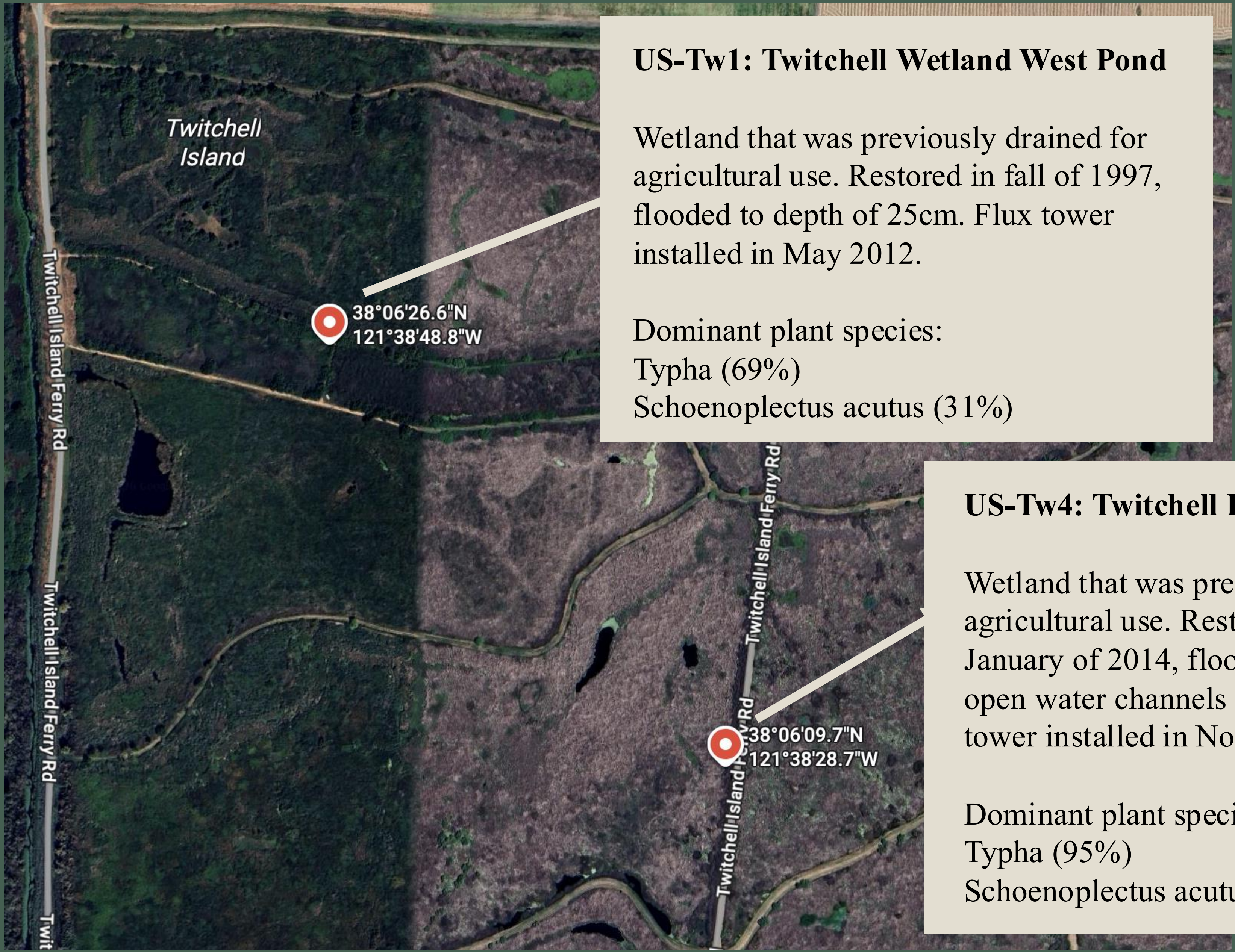
How many years does it take Twitchell East End Wetland, a younger restored wetland, to reach GPP behavior that is similar to Twitchell Wetland West Pond, an older, more established restored wetland?

GPP = Gross Primary Production

Total amount of carbon plants pull out of the atmosphere via photosynthesis over a given time period



Twitchell Island is located in the Sacramento-San Joaquin Delta in California



US-Tw1: Twitchell Wetland West Pond

Wetland that was previously drained for agricultural use. Restored in fall of 1997, flooded to depth of 25cm. Flux tower installed in May 2012.

Dominant plant species:
Typha (69%)
Schoenoplectus acutus (31%)

US-Tw4: Twitchell East End Wetland

Wetland that was previously drained for agricultural use. Restored gradually starting in January of 2014, flooded to depth of 25cm with open water channels at depth of 60cm. Flux tower installed in November 2013.

Dominant plant species:
Typha (95%)
Schoenoplectus acutus(5%)

Objective

Build a predictive model of GPP from HLS remote sensing data, validate its performance using tower-derived AmeriFlux GPP, and apply it across sites to evaluate whether the younger restored wetland approaches the seasonal GPP behavior of the older restored wetland.

Methods

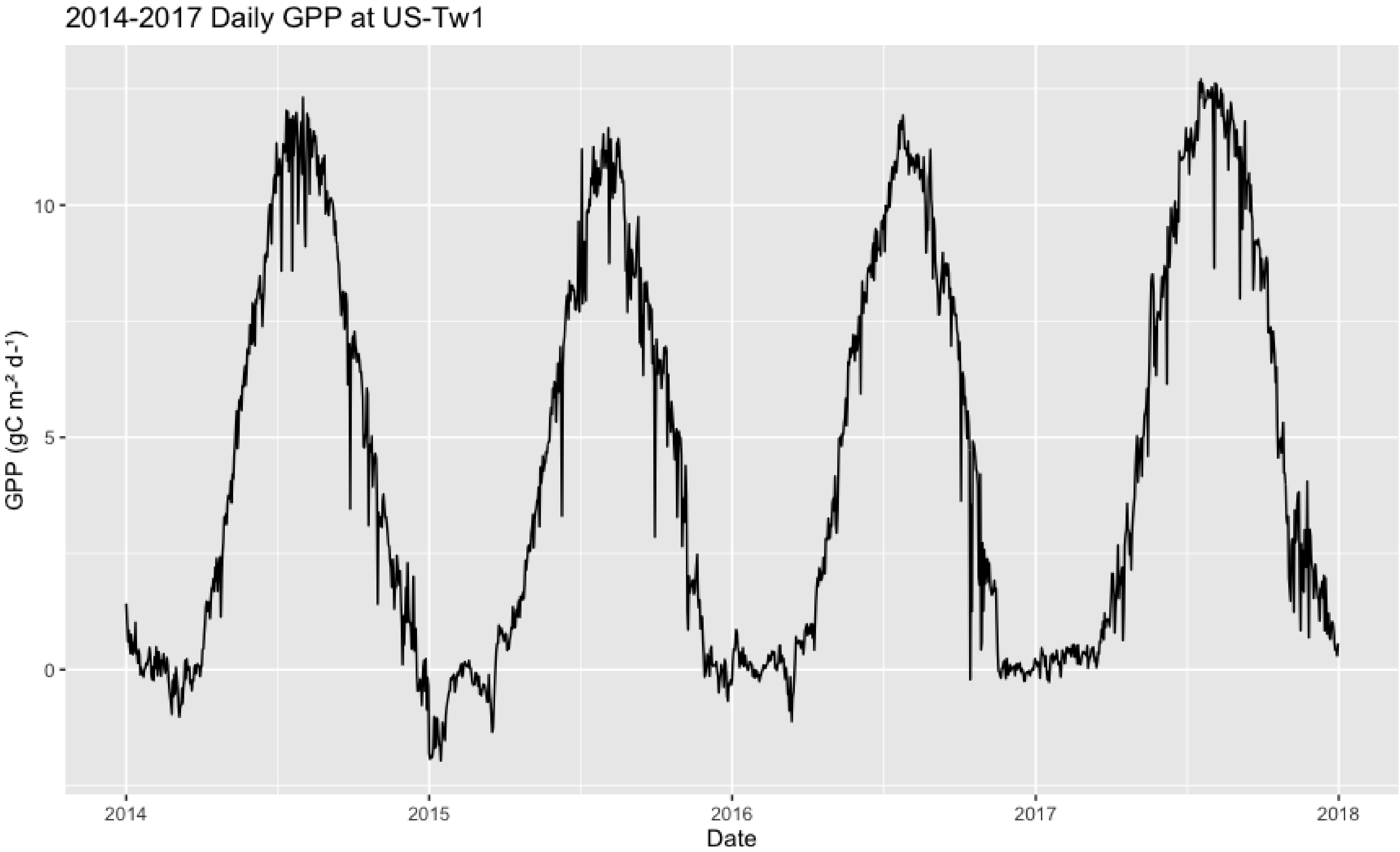
Groundtruth Data: AmeriFlux Daily GPP

Remote Sensing Satellite Data:
HLS - Harmonized Landsat and Sentinel-2

Train:
Tw1: 2016 - 2017
Tw4: 2014 - 2015

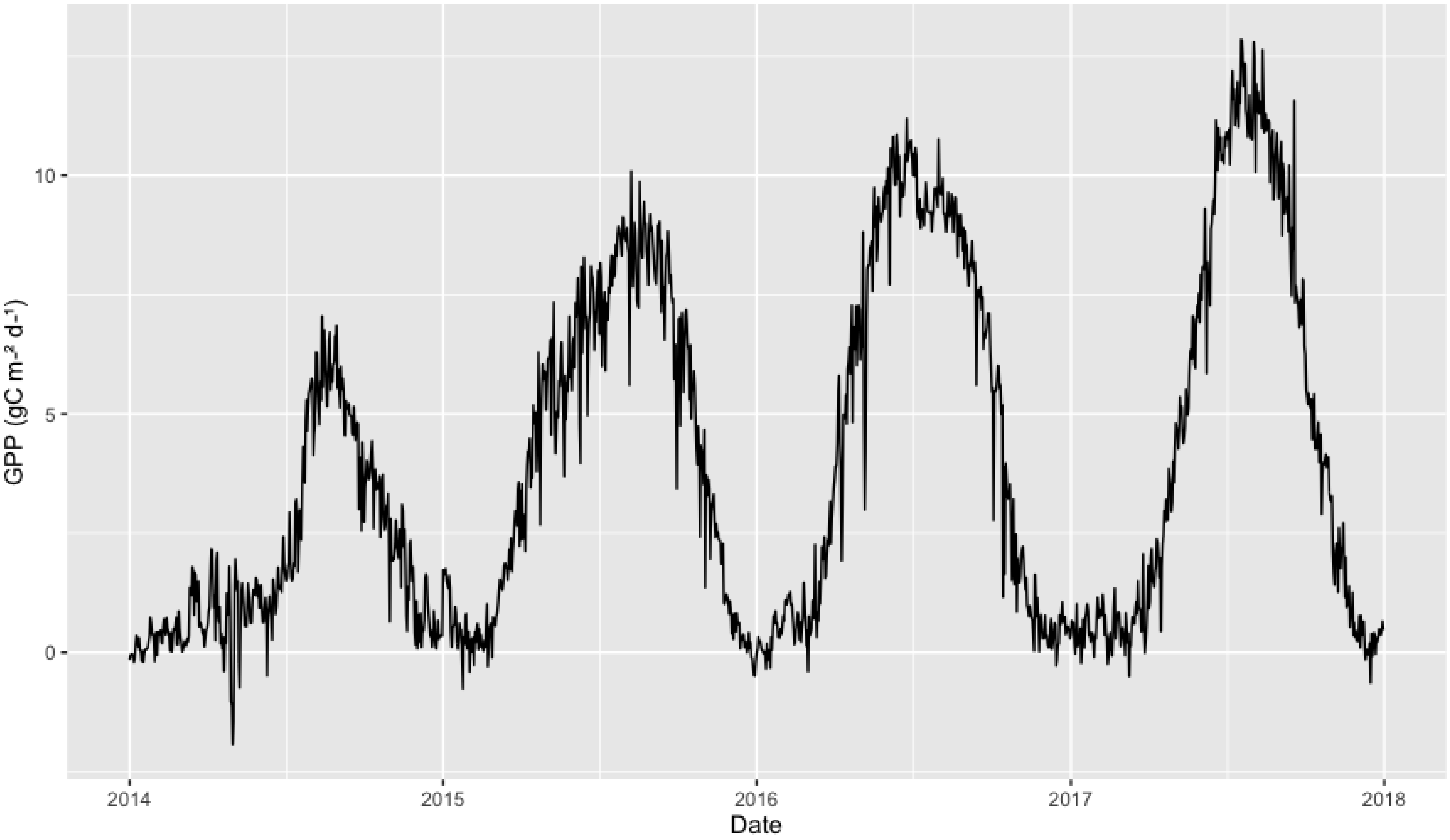
Test:
Tw1: 2014 - 2015
Tw4: 2016 - 2017

Predict: Tw1 and Tw4 2018



Daily GPP at US-Tw1 from 2014-2017. Daily gross primary productivity (GPP; gC m^{-2} d^{-1}) shows strong seasonality, remaining low through winter, increasing rapidly in spring, peaking in mid-summer, and declining through fall and winter. Source: FLUXNET2015 US-Tw1 Twitchell Wetland West Pond, Dataset. https://doi.org/10.18140/FLX/1440111

2014-2017 Daily GPP at US-Tw4



Daily GPP at US-Tw4 from 2014-2017. Daily gross primary productivity (GPP; gC m^{-2} d^{-1}) shows strong seasonality, remaining low through winter, increasing rapidly in spring, peaking in mid-summer, and declining through fall and winter. Source: FLUXNET2015 US-Tw4 Twitchell East End Wetland, Dataset. https://doi.org/10.18140/FLX/1440111

Gathered HLS spectral reflectance data for 2014-2017 at each flux tower site

- filtered for cloud-free and low aerosol days
- merged LS and S2 and standardized band names

Joined AmeriFlux daily GPP data with cleaned HLS spectral reflectance data

Split into training/testing sets based on years

- Training = 71 observations
- Testing = 70 observations

Computed NDVI, NDMI, and EVI from joined dataset

Fit a Gaussian GAM model using EVI and NDMI to predict daily GPP

NDVI – Normalized Difference Vegetation Index

measure of vegetation greenness and density

NDVI = (Band 5 – Band 4) / (Band 5 + Band 4)

EVI – Enhanced Vegetation Index

measure of vegetation greenness with improved performance in dense vegetation and atmospheric interference

EVI = 2.5 * ((Band 5 – Band 4) / (Band 5 + 6 * Band 4 – 7.5 * Band 2 + 1))

NDMI – Normalized Difference Moisture Index

measure of vegetation moisture

NDMI = (Band 5 – Band 6) / (Band 5 + Band 6)

Retrieved four 2018 HLS images each for Tw1 and Tw4

AOI = 500m buffer around each flux tower

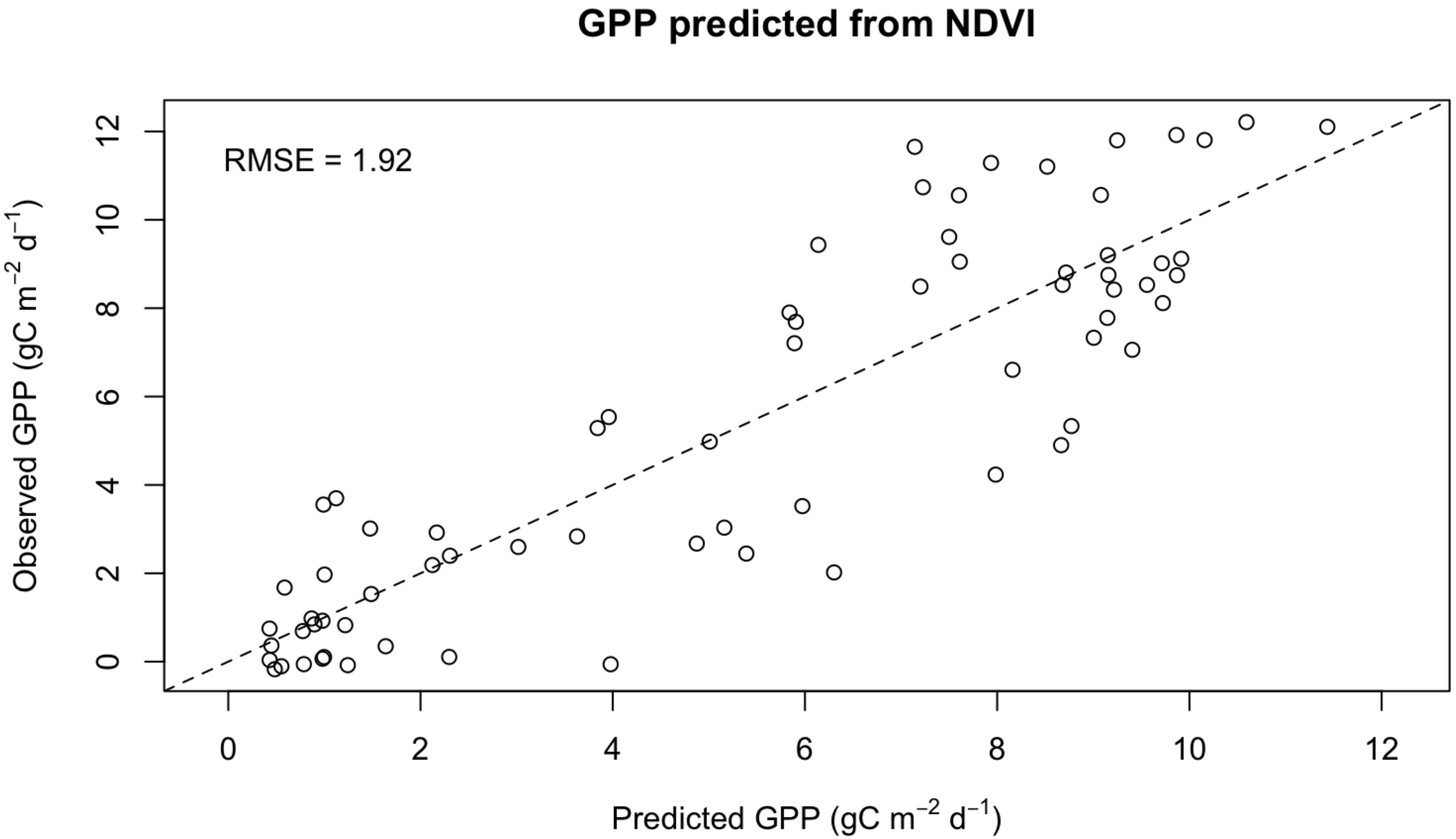
Selected one cloud-free, low aerosol image per season:

- Winter: 01-20-2018
- Spring: 04-20-2018
- Summer: 08-18-2018
- Fall: 11-06-2018

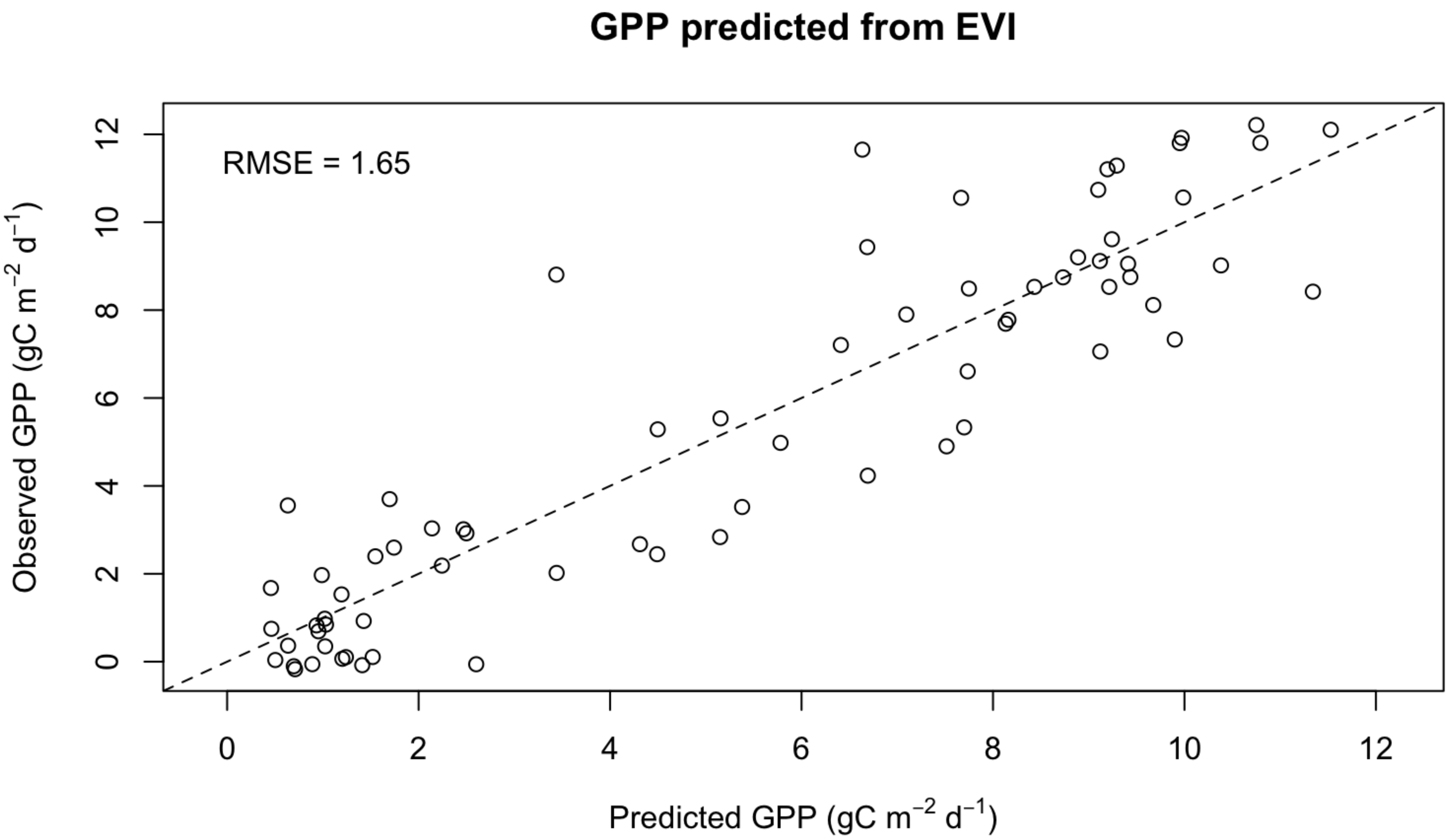
Derived NDMI and EVI from each image, added them as predictor layers, and applied the GAM model to predict GPP across each site

Results

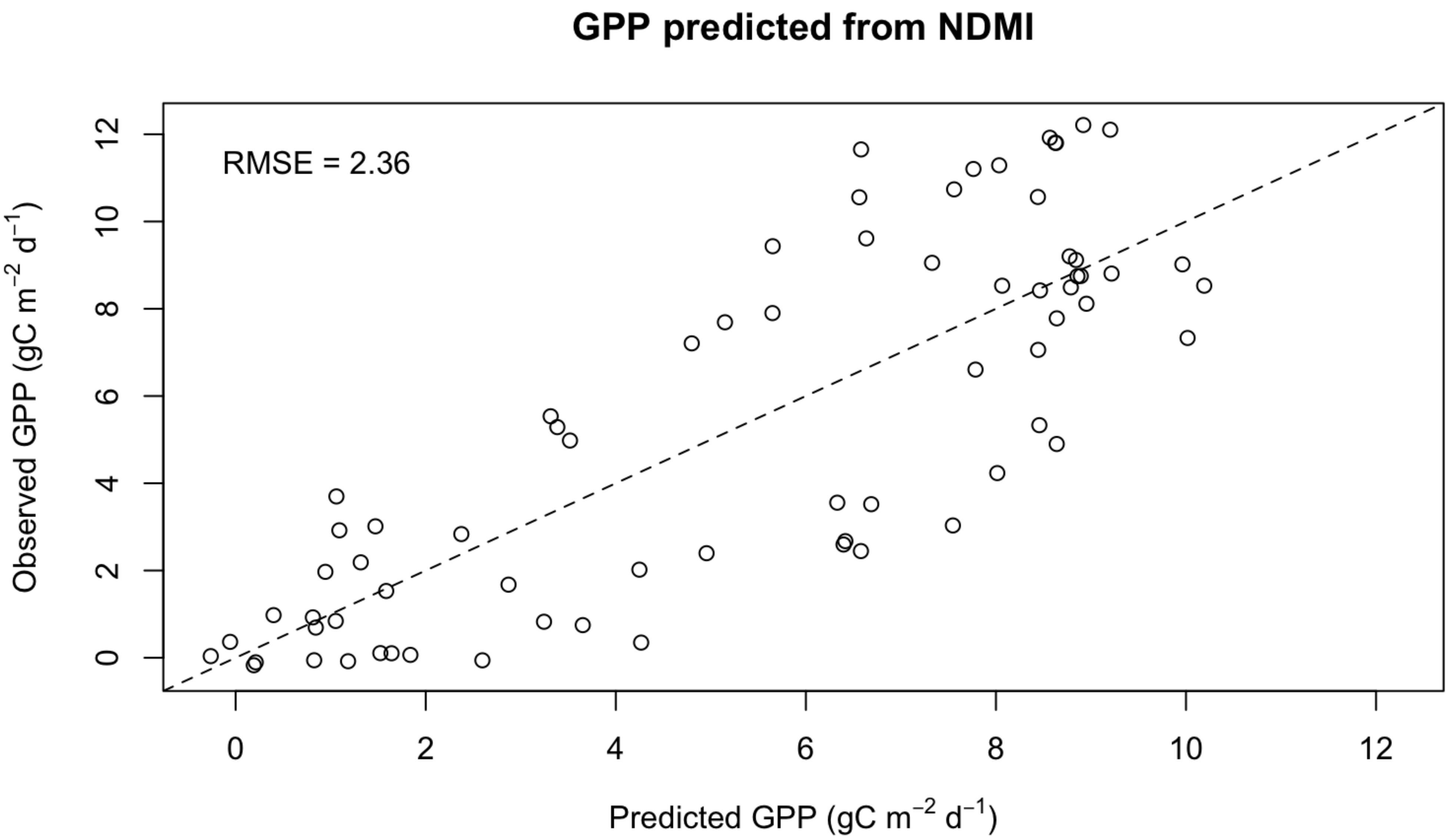
Training the Model



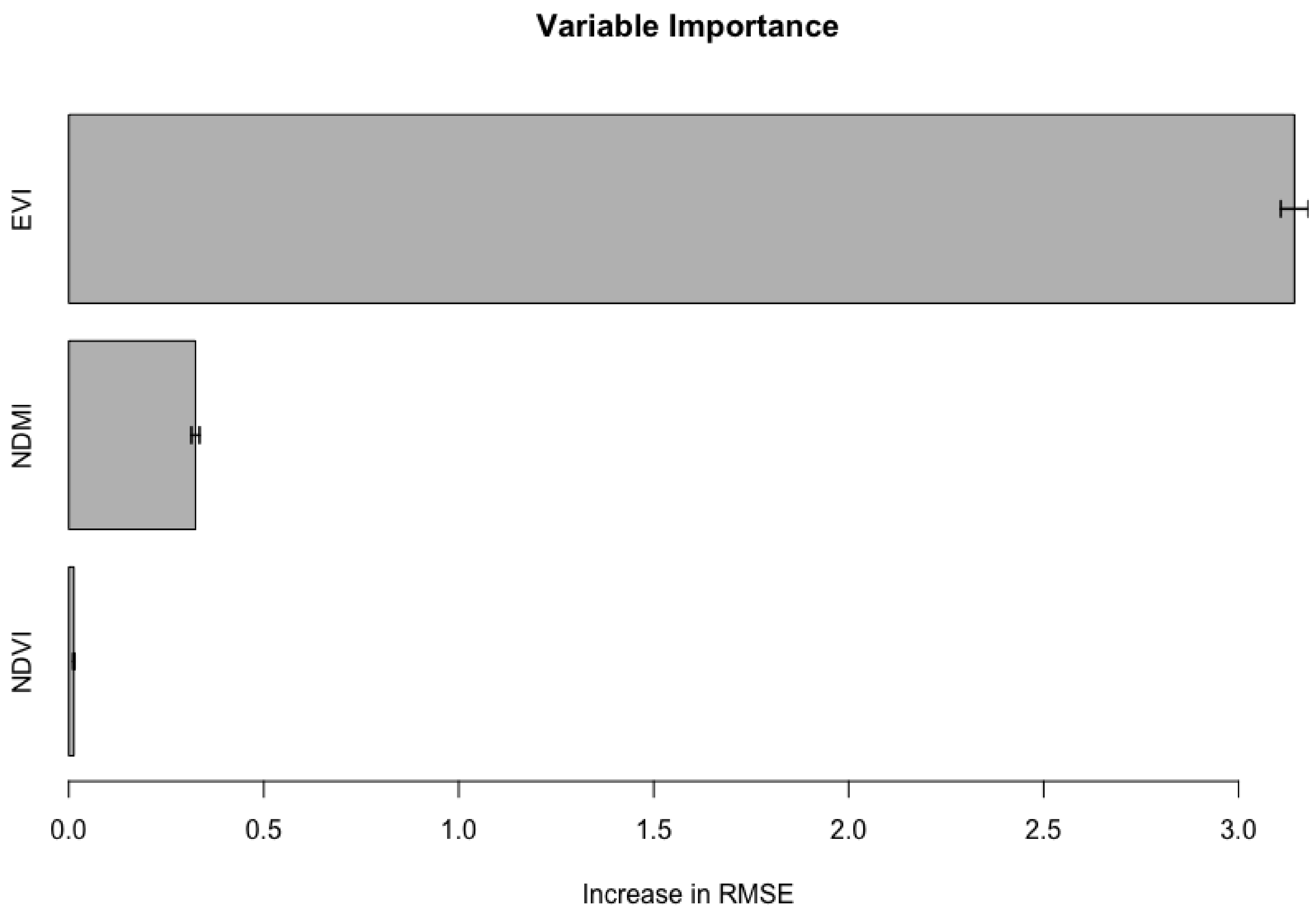
Observed vs. predicted daily GPP using NDVI computed from HLS reflectance.
Training RMSE = 1.92 gC m^{-2} d^{-1}

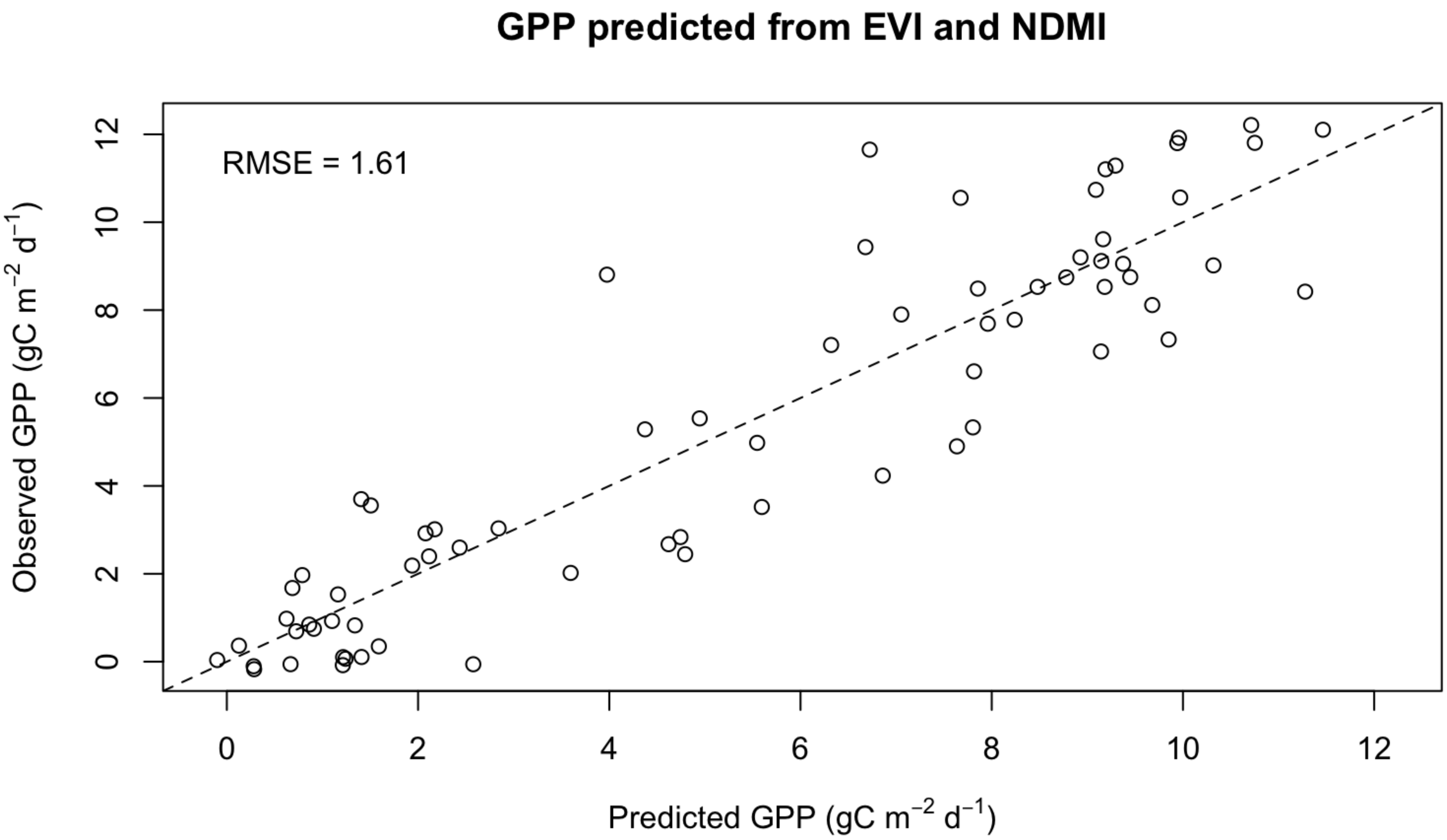


Observed vs. predicted daily GPP using EVI computed from HLS reflectance.
Training RMSE = 1.65 gC m-^{2} d-^{1}



Observed vs. predicted daily GPP using NDMI computed from HLS reflectance.
Training RMSE = 2.36 gC m^{-2} d^{-1}

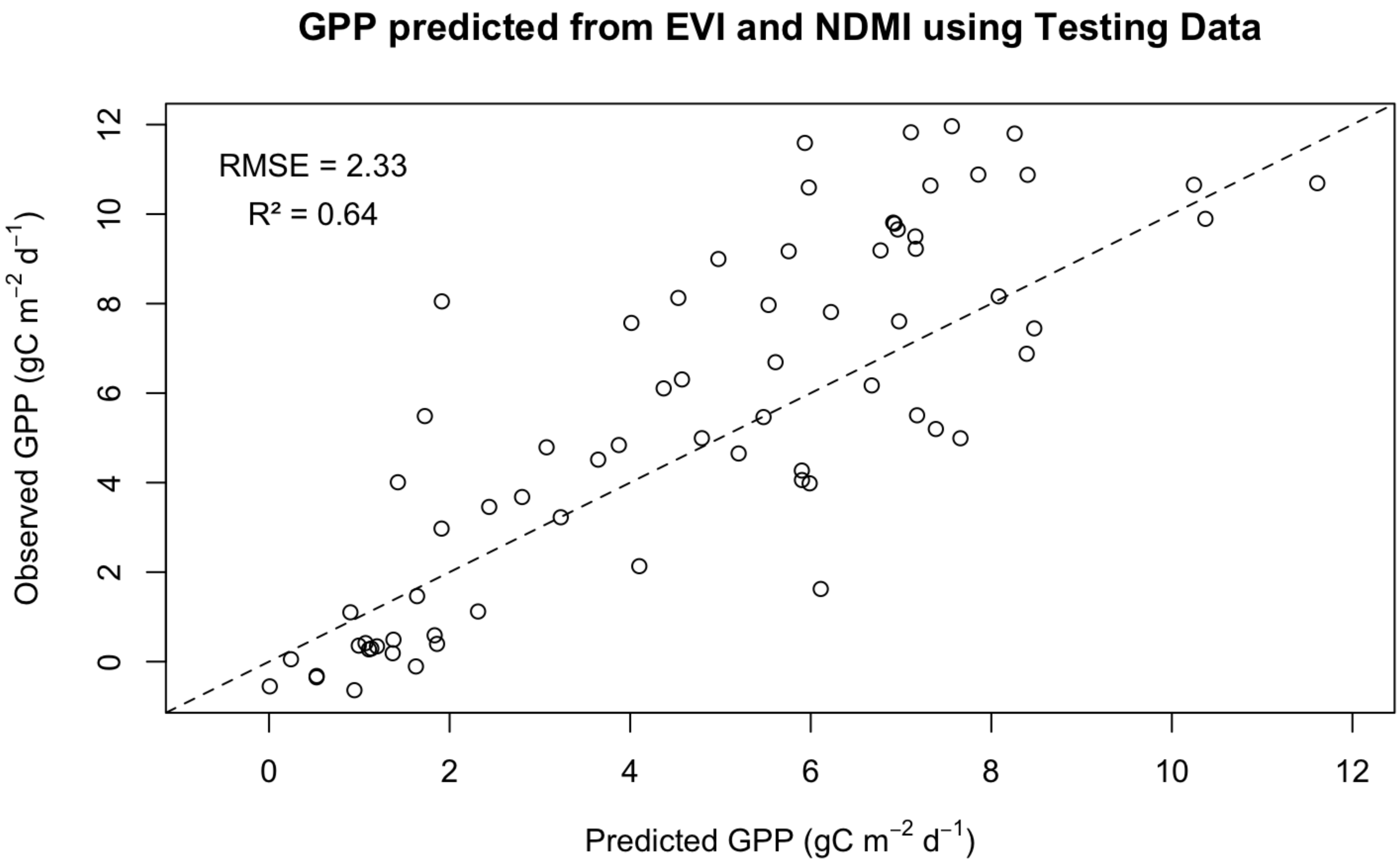

Variable Importance
EVI
NDMI
NDVI
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Increase in RMSE



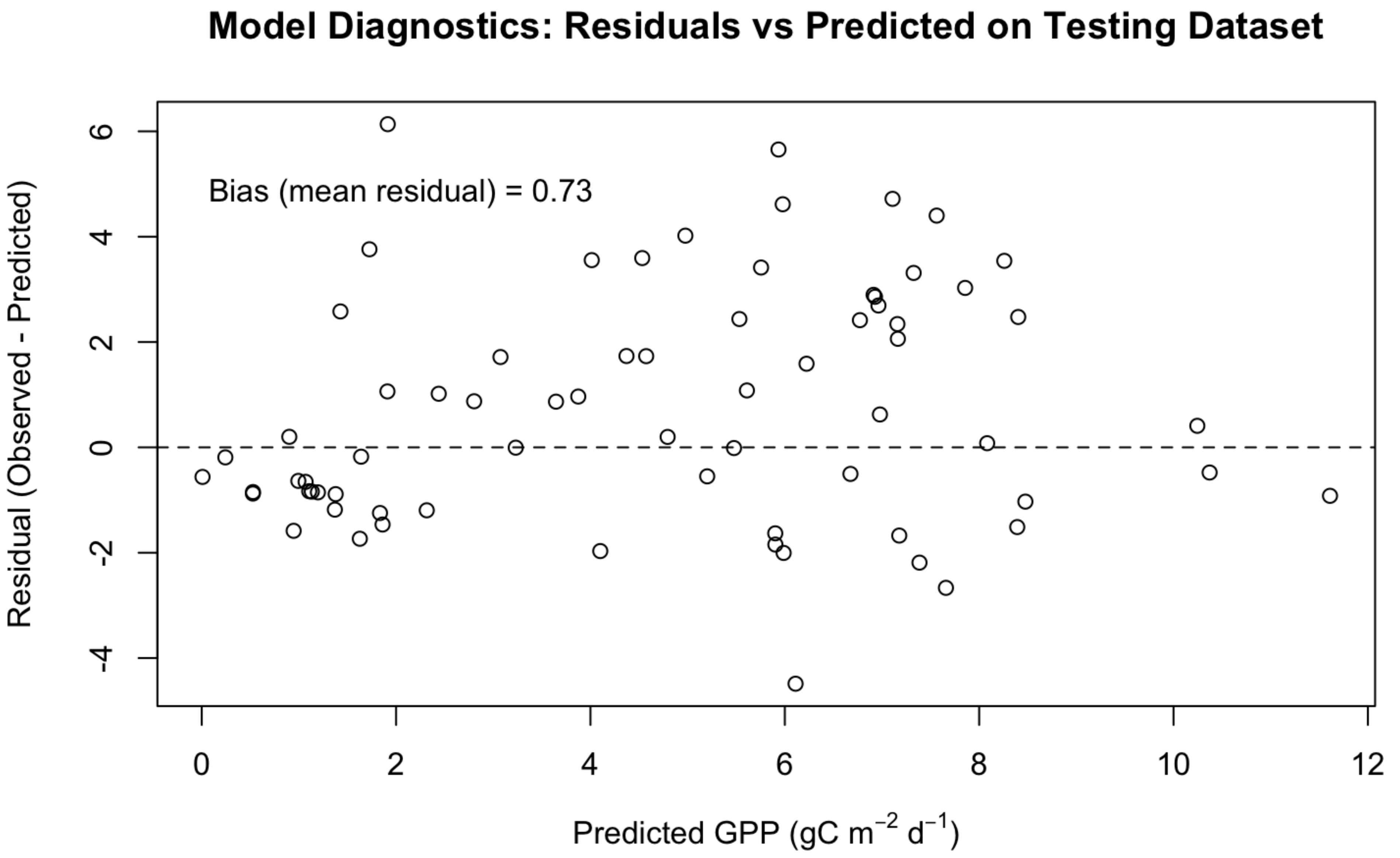
Observed vs. predicted daily GPP using EVI and NDMI computed from HLS reflectance.
Training RMSE = 1.61 gC m^{-2} d^{-1}

Results

Testing the Model



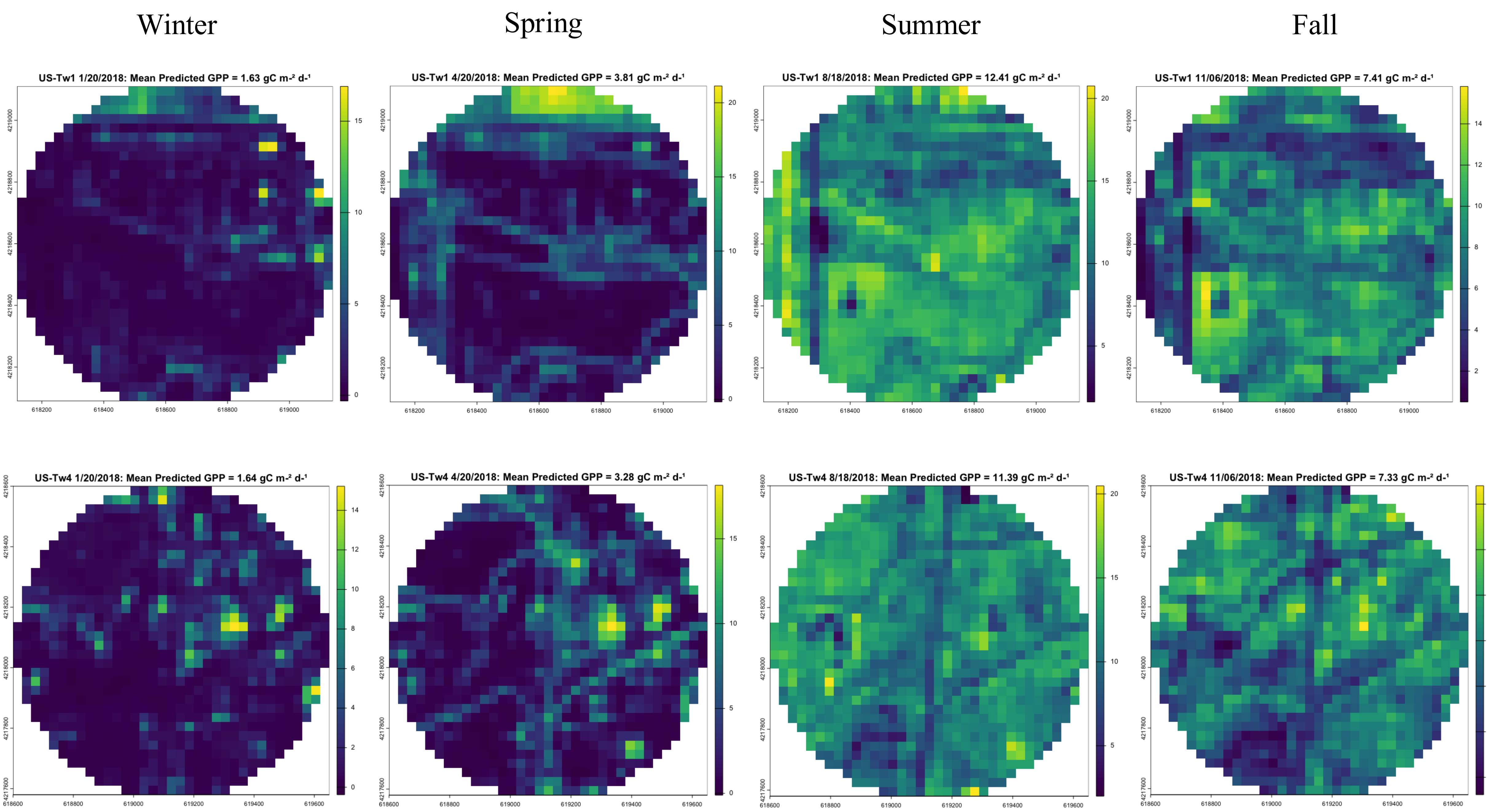
Observed vs. predicted daily GPP on the test dataset using EVI and NDMI computed from HLS reflectance.
Test RMSE = 2.33 gC m^{-2} d^{-1}
Test R^2 = 0.64



On average, observed GPP is 0.73 gC m-² d-¹ higher than the model's predicted GPP.

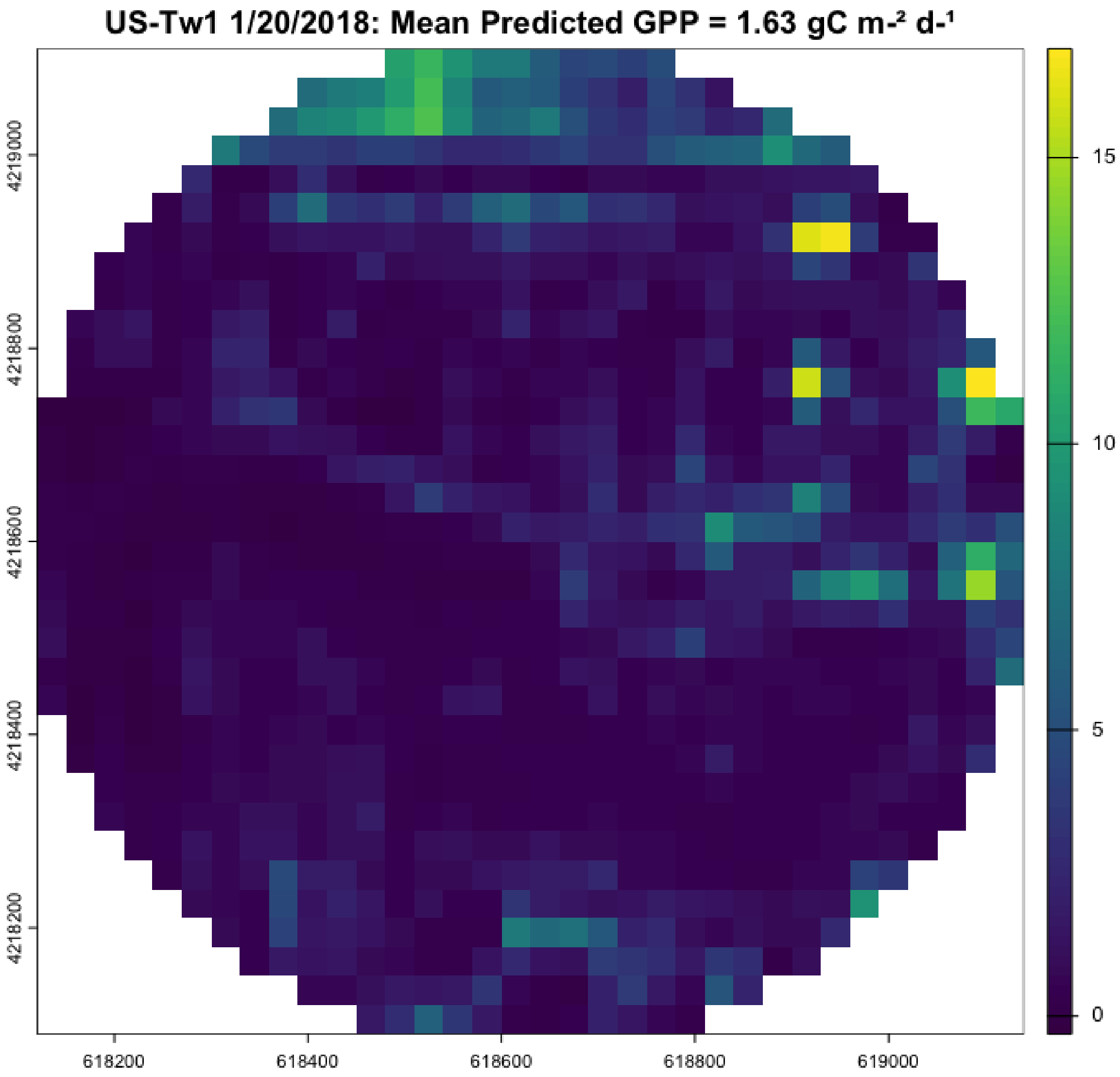
Results

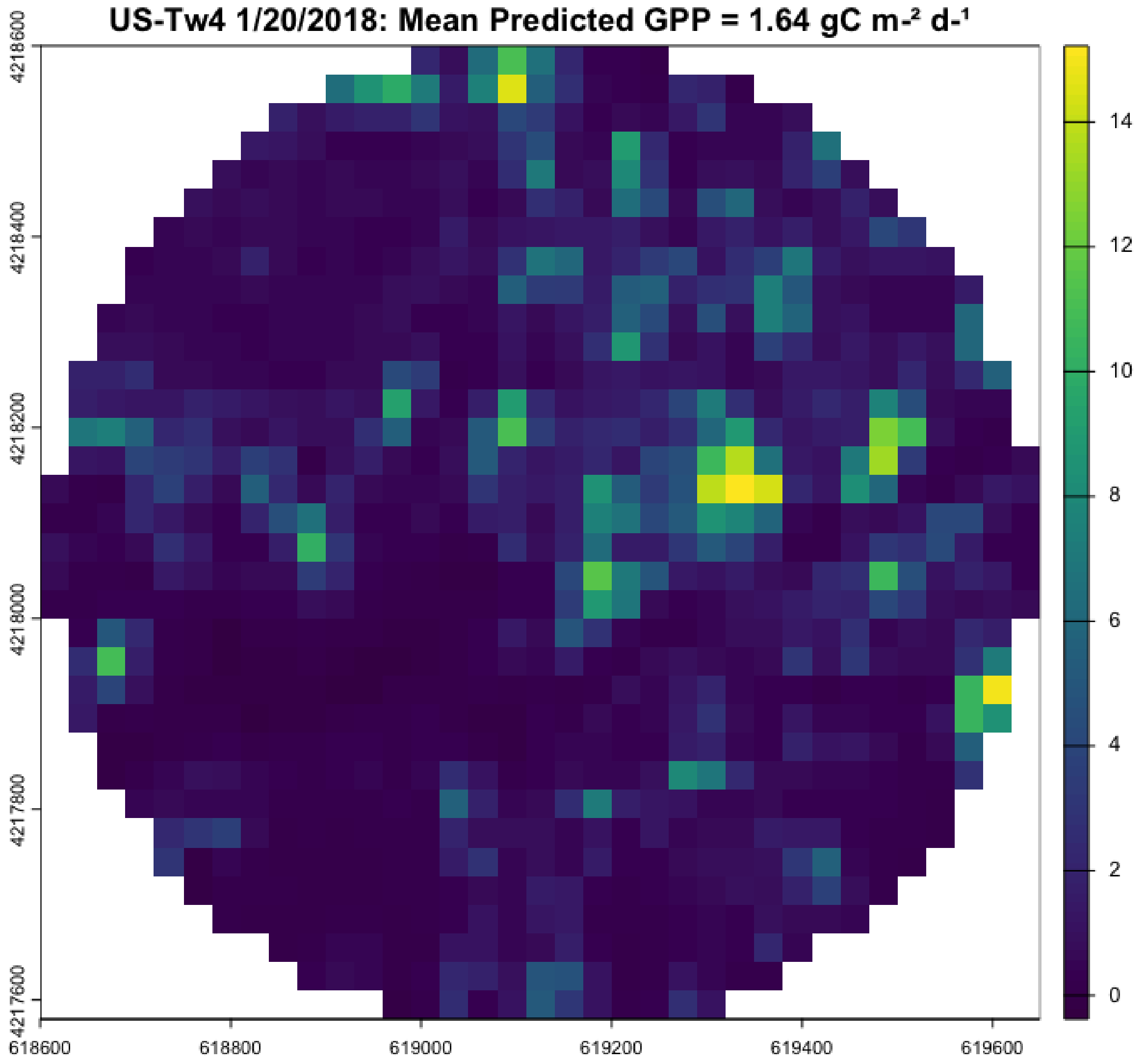
Predicting GPP using Remote Sensing Data



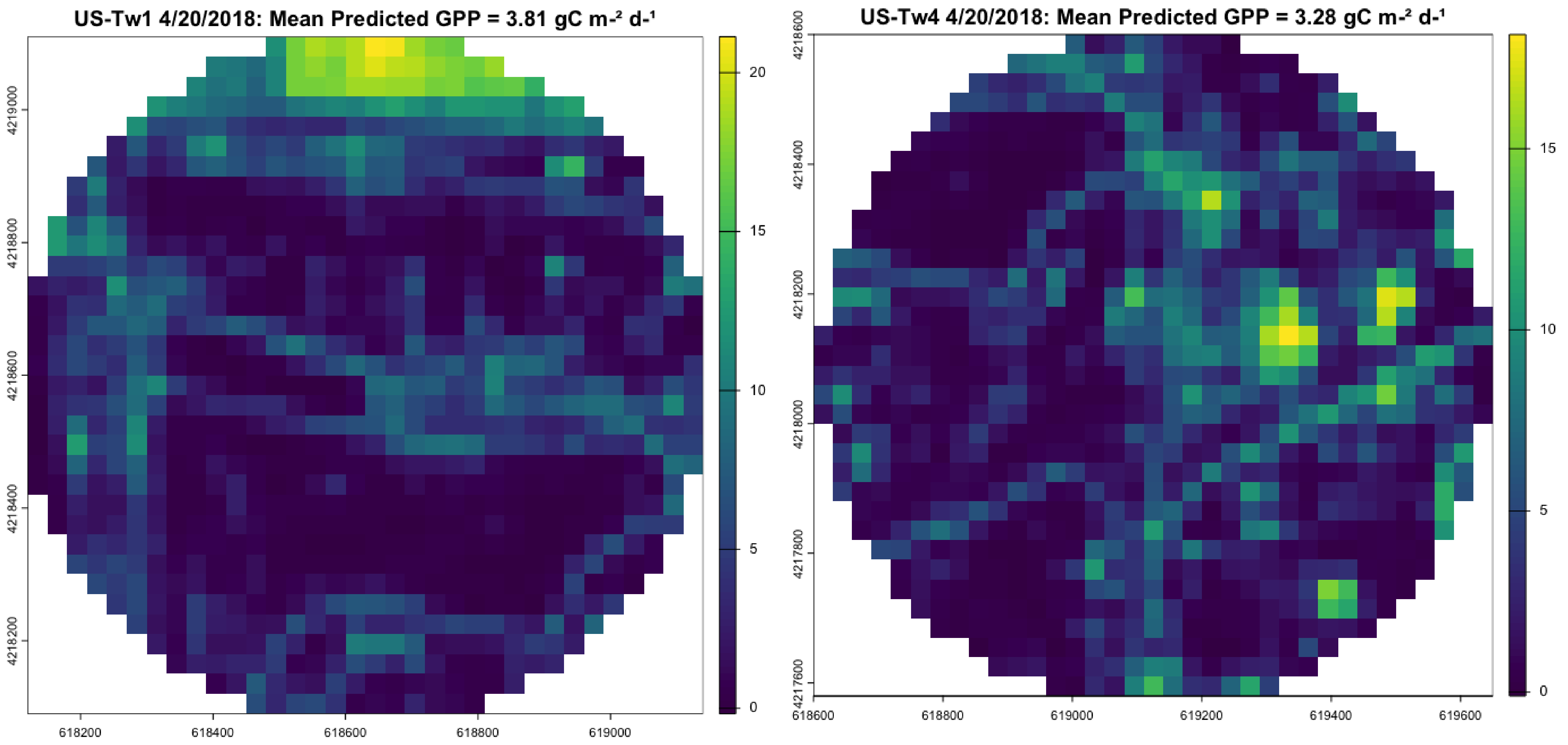
Winter
Spring
Summer
Fall
US-Tw1 1/20/2018: Mean Predicted GPP = 1.63 gC m-² d-¹
US-Tw1 4/20/2018: Mean Predicted GPP = 3.81 gC m-² d-¹
US-Tw1 8/18/2018: Mean Predicted GPP = 12.41 gC m-² d-¹
US-Tw1 11/06/2018: Mean Predicted GPP = 7.41 gC m-² d-¹
US-Tw4 1/20/2018: Mean Predicted GPP = 1.64 gC m-² d-¹
US-Tw4 4/20/2018: Mean Predicted GPP = 3.28 gC m-² d-¹
US-Tw4 8/18/2018: Mean Predicted GPP = 11.39 gC m-² d-¹
US-Tw4 11/06/2018: Mean Predicted GPP = 7.33 gC m-² d-¹

Winter


US-Tw1 1/20/2018: Mean Predicted GPP = 1.63 gC m-² d-¹
4219000
4218800
4218600
4218400
4218200
618200
618400
618600
618800
619000
15
10
5
0

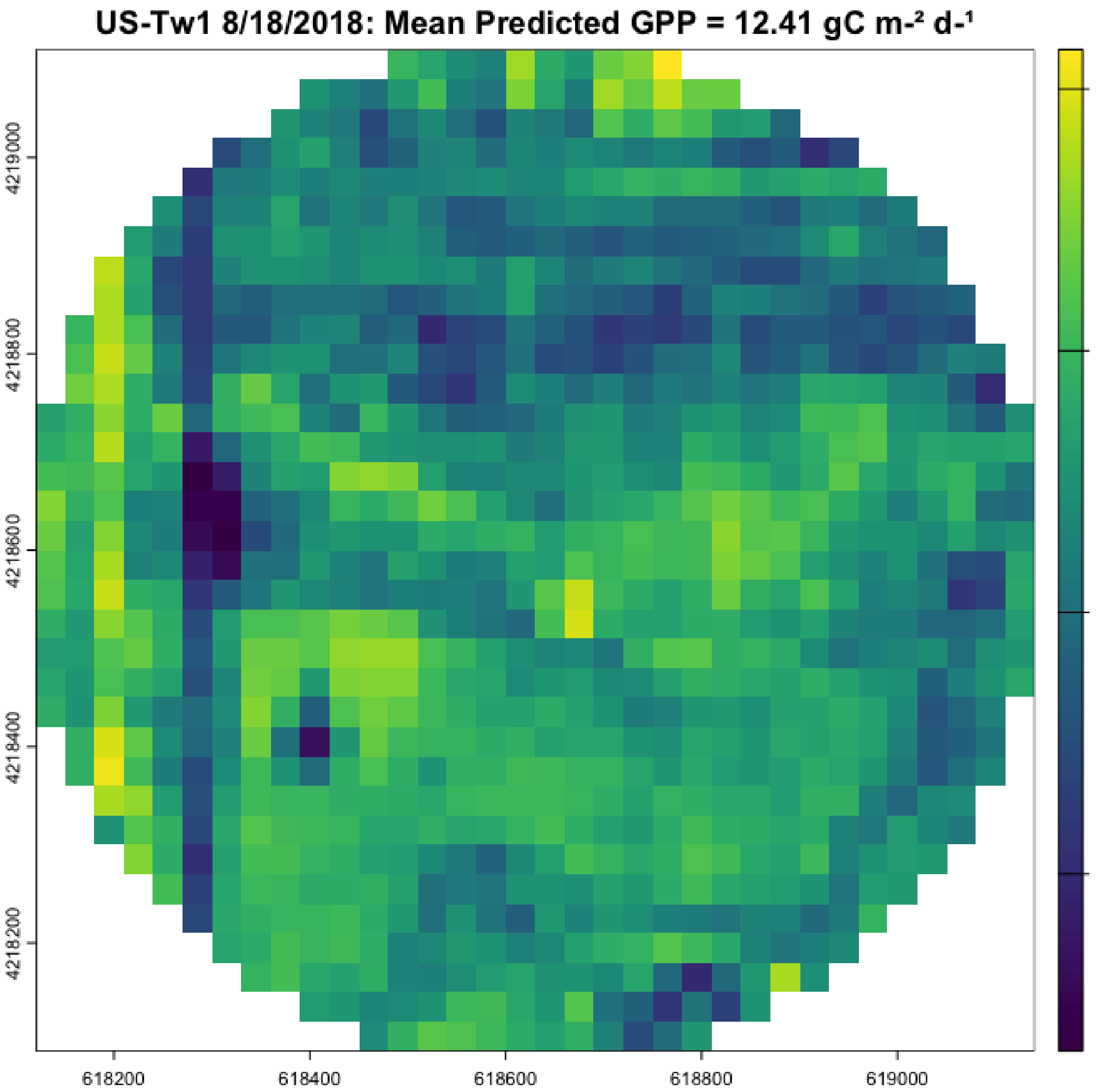

US-Tw4 1/20/2018: Mean Predicted GPP = 1.64 gC m-² d-¹
4218600
4218400
4218200
4218000
4217800
4217600
618600
618800
619000
619200
619400
619600
14
12
10
8
6
4
2
0

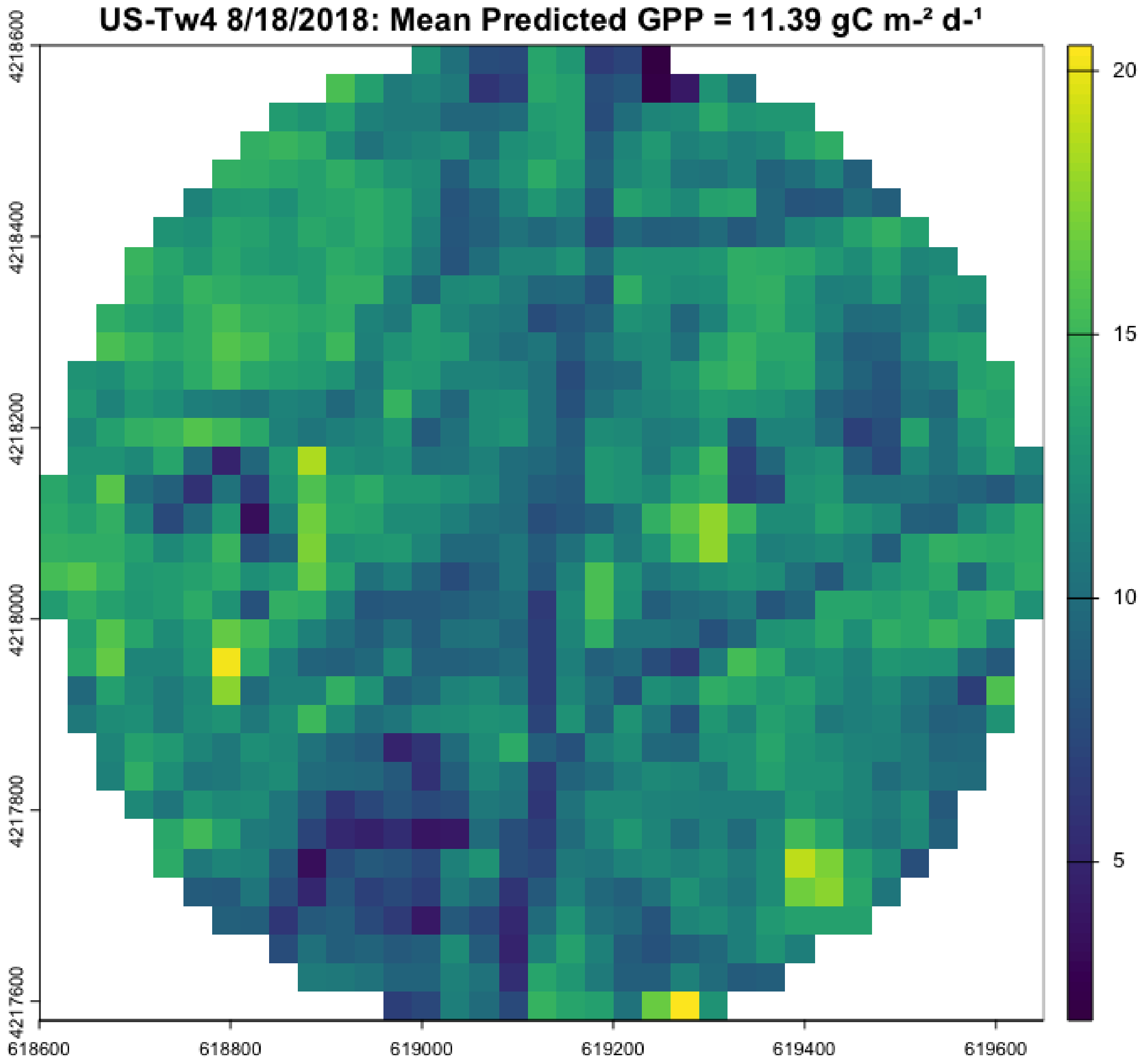
Spring



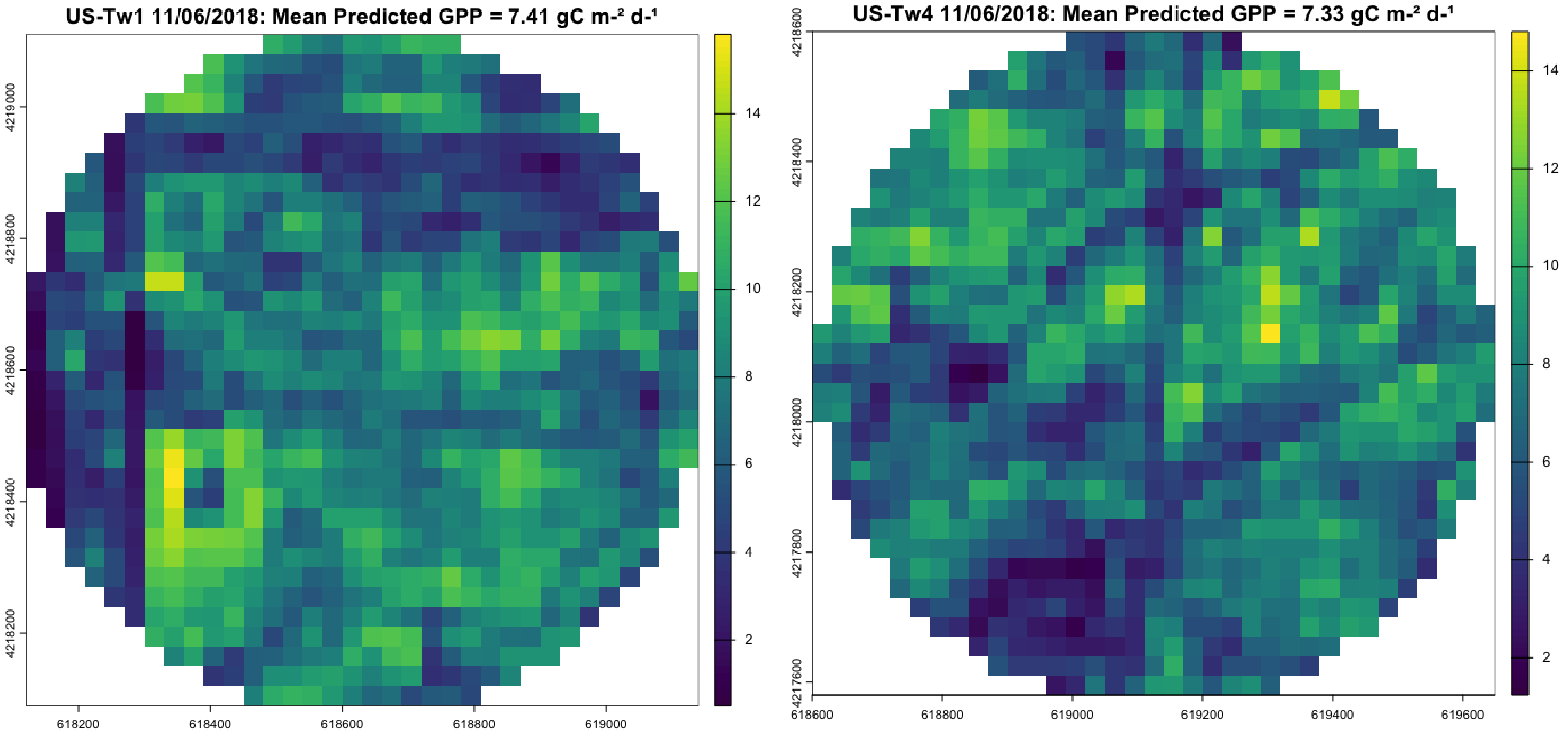
US-Tw1 4/20/2018: Mean Predicted GPP = 3.81 gC m-² d-¹
US-Tw4 4/20/2018: Mean Predicted GPP = 3.28 gC m-² d-¹

Summer


US-Tw1 8/18/2018: Mean Predicted GPP = 12.41 gC m-² d-¹
4219000
4218800
4218600
4218400
4218200
618200
618400
618600
618800
619000


US-Tw4 8/18/2018: Mean Predicted GPP = 11.39 gC m-² d-¹
4218600
4218400
4218200
4218000
4217800
4217600
618600
618800
619000
619200
619400
619600
20
15
10
5

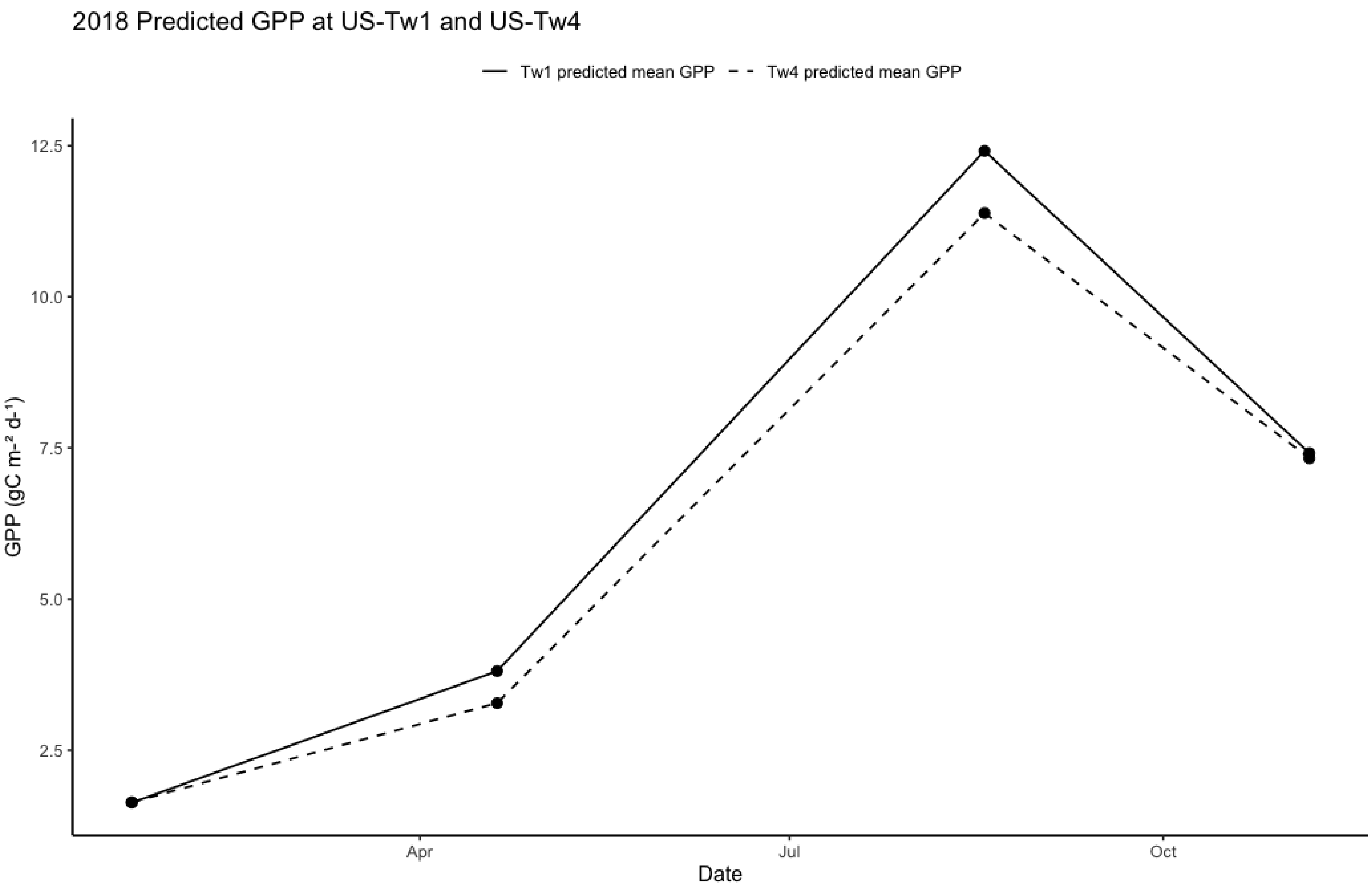
Fall


US-Tw1 11/06/2018: Mean Predicted GPP = 7.41 gC m-² d-¹
US-Tw4 11/06/2018: Mean Predicted GPP = 7.33 gC m-² d-¹
4219000
4218800
4218600
4218400
4218200
618200
618400
618600
618800
619000
2
4
6
8
10
12
14
4218600
4218400
4218200
4218000
4217800
4217600
618600
618800
619000
619200
619400
619600

Conclusion

Four years after restoration, Tw4 has almost approached GPP of Tw1 based on model predictions

The model captured the broad seasonal pattern in GPP across both wetlands



Limitations and Opportunities for Improvement

- Use a flux footprint instead of a simple 500m buffer for AOI
- Filter GPP observations by wind direction to retain only days when tower measurements were likely capturing wetland fluxes
- Performance could likely improve with additional spectral indices such as VARI
- Compare GAM performance with models such as XGBoost

If refined further, a remote sensing approach could help land managers assess whether wetland restoration is delivering anticipated carbon-related benefits.

Questions?

References

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